MF

CASE STUDY

PUBLIC SECTOR / CUSTOMER SUPPORT

NLP Conversation Mining Pipeline

A production NLP pipeline for a Dutch municipality, combining issue extraction, embedding-based clustering and sentiment analysis to turn chatbot conversations into recurring topics and structured FAQs.

KEY METRIC

1,000+

chatbot conversations analyzed

200

Unique FAQs generated

5–6

Major topic clusters surfaced

Manos Fragkiadakis

AI Engineer

PUBLIC SECTOR / CUSTOMER SUPPORT

1,000+ support conversations analyzed.

PROJECT
Conversation Mining Pipeline

CLIENT
Dutch municipality (via Backlight)

INDUSTRY
Public Sector / Customer Support

ROLE
AI Engineer

CONTEXT
A Dutch municipality had deployed a chatbot to help visitors of a major public event, generating more than 1,000 real conversations — but no practical way to see at scale what users were asking, where they were struggling, or what information was missing.

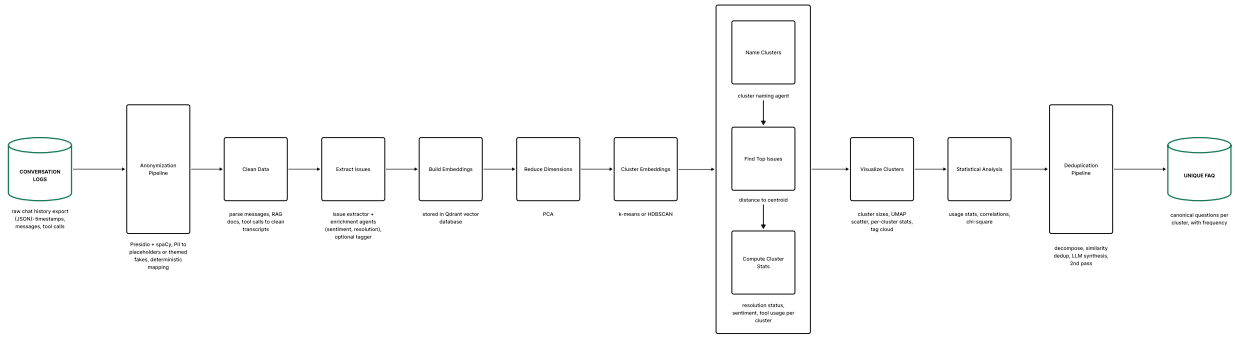
Challenge

Once the chatbot was live, the conversation history became a valuable but difficult-to-use source of product feedback. Questions were repeated in different forms, individual conversations could contain several issues, and manually reviewing more than 1,000 transcripts was not a realistic way to identify recurring issues and knowledge gaps. The team needed a structured view of what users were asking, which problems appeared most often, and where the chatbot or underlying knowledge base could be improved.

Approach

I built a conversation-mining pipeline that cleans and anonymizes the raw chat history and extracts structured user issues from each conversation. The issues are embedded, stored in Qdrant, reduced in dimensionality and clustered into recurring topics with K-Means or HDBSCAN; the clustering also yields each topic's most representative issues, and an agent then names the clusters and provides a short description for easy review. A second stage decomposes compound questions, removes duplicates by vector similarity and produces canonical FAQs. Sentiment, resolution status and tool-use analysis show how the chatbot is performing, and a Streamlit dashboard lets stakeholders explore clusters and inspect the conversations behind them.

Solution Overview



The pipeline from raw conversation logs to a deduplicated FAQ: anonymization, issue extraction, embedding, clustering, cluster naming and statistics, then deduplication.

How It Works

- Issue extraction
- Embedding clustering
- Topic analysis
- Sentiment analysis
- User analytics

Tech Stack

- LangGraph
- Vertex AI (Gemini)
- Qdrant
- scikit-learn
- Streamlit

Results & Impact

The pipeline analyzed more than 1,000 real conversations, surfaced 5–6 major topic clusters and generated 200 unique FAQs with suggestions for improving the knowledge base. Sentiment, resolution and tool-use analysis also exposed where users were becoming frustrated and where chatbot behavior needed attention. Instead of manually reading conversation logs, the team had a structured view of recurring demand and concrete material for improving chatbot and FAQ content.

1,000+
Conversations analyzed

200
Unique FAQs generated

5–6
Major topic clusters surfaced

Project Details

LOCATION
Netherlands

TIMELINE
Q4 2025

AGENT PLATFORM
Google Cloud Vertex AI

TEAM SIZE
Built by me within Backlight's AI team

ENGAGEMENT MODE
AI Engineer at Backlight

ABOUT MANOS

Manos Fragkiadakis is an Applied AI Engineer building document intelligence and AI knowledge systems. He has over five years of Python software development experience and has built custom ERP and CRM systems for professionals. His recent work includes cited RAG across 3,000+ page legal case files and court-monitoring automation software used by 10+ law offices. He previously worked as an AI engineer and taught Natural Language Processing at Utrecht University.

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